2.3 集合的积与无交并

定理（无交并的"外在"版本的泛性质刻画）$(A_i)_{i\in I}$ 是任意的一族集合，

A 是一个集合，$(\iota_i : A_i \longrightarrow A)_{i\in I}$ 是一族映射，满足：

* 每个 ι_i 都是单射，
* $A = \bigcup\limits_{i\in I} \mathrm{im}(\iota_i)$。
* 对 $\forall i,j\in I$，$i\neq j$，有：$\mathrm{im}(\iota_i)\cap \mathrm{im}(\iota_j)=\varnothing$。

A' 是另一个集合，$(\iota'_i : A_i \longrightarrow A')_{i\in I}$ 是另一族映射，满足：

* 每个 ι'_i 都是单射，
* $A' = \bigcup\limits_{i\in I} \mathrm{im}(\iota'_i)$
* 对 $\forall i,j\in I$，$i\neq j$，有：$\mathrm{im}(\iota'_i)\cap \mathrm{im}(\iota'_j)=\varnothing$。

则有：存在唯一的映射 $\varphi : A \longrightarrow A'$，使得对 $\forall i\in I$，下图交换：

$$\begin{tikzcd}
A \arrow[rr, "\varphi"] & & A' \\
& A_i \arrow[ul, "\iota_i"] \arrow[ur, "\iota'_i"'] &
\end{tikzcd}$$

且映射 φ 是双射。

Proof: 对 $\forall x \in A$. 我们按照下面的方式来定义映射 $\varphi: A \longrightarrow A'$.

$\because x \in A = \bigcup\limits_{i \in I} \mathrm{im}(l_i)$ $\quad \therefore \exists t \in I$, s.t. $x \in \mathrm{im}(l_t)$

$\because l_t$ 是 $A_t \longrightarrow A$ 的映射 $\quad \therefore \exists \alpha \in A_t$, s.t. $l_t(\alpha) = x$

$\because l'_t$ 是 $A_t \longrightarrow A'$ 的映射 $\quad \therefore l'_t(\alpha) \in A'$.

定义 $\varphi(x) := l'_t(\alpha) \in A'$. 显然有 $\varphi(A) \subseteq A'$.

对 $\forall x_1, x_2 \in A$. 有:

$\because x_1 \in A = \bigcup\limits_{i \in I} \mathrm{im}(l_i)$ $\quad \therefore \exists t_1 \in I$, s.t. $x_1 \in \mathrm{im}(l_{t_1})$

$\because l_{t_1}$ 是 $A_{t_1} \longrightarrow A$ 的映射 $\quad \therefore \exists \alpha_1 \in A_{t_1}$, s.t. $l_{t_1}(\alpha_1) = x_1$

$\because l'_{t_1}$ 是 $A_{t_1} \longrightarrow A'$ 的映射 $\quad \therefore l'_{t_1}(\alpha_1) \in A'$.

$\therefore \varphi(x_1) = l'_{t_1}(\alpha_1) \in A'$

$\because x_2 \in A = \bigcup\limits_{i \in I} \mathrm{im}(l_i)$ $\quad \therefore \exists t_2 \in I$, s.t. $x_2 \in \mathrm{im}(l_{t_2})$

$\because l_{t_2}$ 是 $A_{t_2} \longrightarrow A$ 的映射 $\quad \therefore \exists \alpha_2 \in A_{t_2}$, s.t. $l_{t_2}(\alpha_2) = x_2$

$\because l'_{t_2}$ 是 $A_{t_2} \longrightarrow A'$ 的映射 $\quad \therefore l'_{t_2}(\alpha_2) \in A'$

$\therefore \varphi(x_2) = l'_{t_2}(\alpha_2) \in A'$.

若 $x_1 = x_2$, 则有: $x_1 \in \mathrm{im}(l_{t_1})$, $x_1 = x_2 \in \mathrm{im}(l_{t_2})$

$\therefore x_1 \in \mathrm{im}(l_{t_1}) \cap \mathrm{im}(l_{t_2})$ $\quad \therefore t_1 = t_2$

$\therefore l_{t_1}(\alpha_1) = x_1 = x_2 = l_{t_2}(\alpha_2) = l_{t_1}(\alpha_2)$ $\quad \because l_{t_1}$ 是单射 $\quad \therefore \alpha_1 = \alpha_2$

$\therefore \varphi(x_1) = l'_{t_1}(\alpha_1) = l'_{t_2}(\alpha_1) = l'_{t_2}(\alpha_2) = \varphi(x_2)$ $\quad \therefore \varphi$ 是 $A \to A'$ 的映射.

若 $\varphi(x_1)=\varphi(x_2)$，则有：$l'_{t_1}(\alpha_1)=l'_{t_2}(\alpha_2)$

$\because l'_{t_1}(\alpha_1)\in \mathrm{im}(l'_{t_1})$， $l'_{t_1}(\alpha_1)=l'_{t_2}(\alpha_2)\in \mathrm{im}(l'_{t_2})$

$\therefore l'_{t_1}(\alpha_1)\in \mathrm{im}(l'_{t_1})\cap \mathrm{im}(l'_{t_2})$ $\quad\therefore t_1=t_2$

$\therefore l'_{t_1}(\alpha_1)=l'_{t_2}(\alpha_2)=l'_{t_1}(\alpha_2)$ $\quad\because l'_{t_1}$是单射 $\quad\therefore \alpha_1=\alpha_2$

$\therefore x_1=l_{t_1}(\alpha_1)=l_{t_2}(\alpha_1)=l_{t_2}(\alpha_2)=x_2$ $\quad\therefore \varphi$是$A\longrightarrow A'$的单射.

对$\forall \delta\in A'$， $\because \delta\in A'=\bigcup\limits_{i\in I}\mathrm{im}(l'_i)$ $\quad\therefore \exists k\in I$，s.t. $\delta\in\mathrm{im}(l'_k)$

$\because l'_k$是$A_k\longrightarrow A'$的映射 $\quad\therefore \exists\gamma\in A_k$， s.t. $l'_k(\gamma)=\delta$

$\because l_k$是$A_k\longrightarrow A$的映射 $\quad\therefore l_k(\gamma)\in A$

$\therefore \varphi(l_k(\gamma))=l'_k(\gamma)=\delta$ $\quad\therefore \varphi$是$A\longrightarrow A'$的满射.

$\therefore \varphi$是$A\longrightarrow A'$的双射.

对$\forall i\in I$. $\quad\because l_i$是$A_i\longrightarrow A$的映射， φ是$A\longrightarrow A'$的映射

$\therefore \varphi\circ l_i$是$A_i\longrightarrow A'$的映射.

$\because i\in I$ $\quad\therefore l'_i$是$A_i\longrightarrow A'$的映射.

对$\forall\varepsilon\in A_i$，有：$(\varphi\circ l_i)(\varepsilon)=\varphi(l_i(\varepsilon))=l'_i(\varepsilon)$

$\therefore \varphi\circ l_i=l'_i$ $\quad\therefore$ 下图交换：

$$\begin{tikzcd} A \arrow[rr, "\varphi"] & & A' \\ & A_i \arrow[ul, "l_i"] \arrow[ur, "l'_i"'] & \end{tikzcd}$$

$\therefore$存在性得证.

下证唯一性. 假设存在两个映射 $\varphi_1 : A \longrightarrow A'$, $\varphi_2 : A \longrightarrow A'$,

使得: 对 $\forall i \in I$, 下图交换:

①:
$$\begin{tikzcd} A \arrow[rr, "\varphi_1"] & & A' \\ & A_i \arrow[ul, "l_i"] \arrow[ur, "l_i'"'] & \end{tikzcd}$$

②:
$$\begin{tikzcd} A \arrow[rr, "\varphi_2"] & & A' \\ & A_i \arrow[ul, "l_i"] \arrow[ur, "l_i'"'] & \end{tikzcd}$$

对 $\forall \theta \in A$. $\because \theta \in A = \bigcup\limits_{i \in I} \mathrm{im}(l_i)$ $\therefore \exists q \in I$, s.t. $\theta \in \mathrm{im}(l_q)$

$\because l_q$ 是 $A_q \longrightarrow A$ 的映射 $\therefore \exists \mu \in A_q$, s.t. $l_q(\mu) = \theta$

$$\therefore \varphi_1(\theta) = \varphi_1(l_q(\mu)) = (\varphi_1 \circ l_q)(\mu) \underset{\text{图①交换}}{=} l_q'(\mu)$$

$$\underset{\text{图②交换}}{=} (\varphi_2 \circ l_q)(\mu) = \varphi_2(l_q(\mu)) = \varphi_2(\theta)$$

$\therefore \varphi_1 = \varphi_2$ $\therefore$ 唯一性得证 $\square$